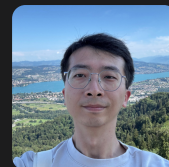


Yuan YIN 银元

AI Researcher · Model adaptation, learned dynamics, autonomous driving

 yuan-yin.github.io ·  Google Scholar ·  yuan-yin-nn ·  yuan-yin ·  Paris, France



Profile

AI researcher working on **parameter-efficient adaptation of pretrained vision models** and **learned physical dynamics**, applied to autonomous driving. My PhD established hybrid physics/deep-learning modelling and out-of-distribution generalisation for dynamical systems; I now transfer that work to autonomous driving at an open industrial research lab, where results are published rather than kept internal. I favour methods that exploit the structure of a problem — physical priors, geometry, pretrained representations — and I work on integrating deep learning into existing non-ML systems.

Experience

Valeo.ai *Paris, France*

► **AI Researcher** *Dec 2024 – Present*

Perception, motion forecasting and decision making for autonomous driving; adapting large pretrained models and learning at scale

► **Postdoctoral AI Researcher** *Apr 2024 – Nov 2024*

Corner-case generation for learning robust self-driving cars

Sorbonne Université ISIR, MLIA *Paris, France*

► **Postdoctoral Researcher** *Jul 2023 – Dec 2023*

Supervised ongoing research projects; authored a tutorial on physics-aware deep learning

► **PhD Student & Teaching Assistant** *Oct 2019 – Jun 2023*

Advisors: Patrick Gallinari & Nicolas Baskiotis
Physics-aware deep learning for dynamical systems: hybrid physics/DL modelling, out-of-distribution generalisation, continuous-time dynamics

► **Research Intern, Deep Learning** *Feb 2019 – Sep 2019*

Imputing spatiotemporal data with generative models

Inria Paris Research Intern, NLP *2018*

Beihang Univ. Research Intern, Vision *2015–16*

Education

Sorbonne Université fka UPMC (Paris-6) *Paris, France*

► **PhD, Machine Learning & Deep Learning** *2023*

► **MSc2 DAC Master Data Science Paris** *2019*

Université Paris Cité fka Paris-Diderot (Paris-7) *Paris, France*

► **MSc1 MPRI Parisian Research Master in CS** *2018*

► Univ. Diploma, French Language & Civilization *2017*

Beihang University *Beijing, China*

► **BSc, Computer Science** *2016*

Technical Skills

ML / DL PyTorch, JAX, Hugging Face, multi-GPU & multi-node training

Coding Infra Python, C/C++, Java, OCaml, Matlab, $\text{\LaTeX}$
Linux, Slurm, Docker, Git, Weights & Biases

Languages

French Bilingual *last exam ▶ C1 (2017)*

English Full Professional *last exam ▶ B2 (2015)*

Mandarin Native

Awards & Honours

Accessit for the 2024 AI Thesis Prize of the **French Association for Artificial Intelligence (AFIA)**

Best Paper & Oral, NeurIPS 2025 CCFM Workshop

Top Reviewer, NeurIPS 2023

Service & Teaching

Conference reviewing — NeurIPS 2021–26, ICLR 2023–26, ICML 2022–26, CVPR 2025–26, ICRA 2026, ECML-PKDD 2021, ACM-MM 2021

Workshop reviewing — CCFM @ NeurIPS 2025, ML4PS @ NeurIPS 2022–24, Physics4ML @ ICLR 2023, SynS&ML @ ICML 2023, ROAM @ ECCV 2024

Teaching — Sorbonne Université, Engineering Dept. (UFR 919), 2019–22, in French. Undergraduate: C programming (L1), algorithmics (L2), probability (L3). Graduate: ML research methodology (M2)

Selected Publications

Conference & journal papers ** equal contribution*

· E. Kirby, A. Boulch, Y.-H. Xu, **Y. Yin**, G. Puy, É. Zablocki, A. Bursuc, S. Gidaris, R. Marlet, F. Bartoccioni, A.-Q. Cao, N. Samet, T.-H. Vu, and M. Cord. Driving on registers. In **CVPR 2026**.

· Y.-H. Xu*, **Y. Yin***, T.-H. Vu, A. Boulch, É. Zablocki, and M. Cord. PPT:

Pre-training with pseudo-labeled trajectories for motion forecasting. In *ICRA 2026*.

- **Y. Yin**, S. Venkataramanan, T.-H. Vu, A. Bursuc, and M. Cord. IPA: An information-reconstructive input projection framework for efficient foundation model adaptation. *TMLR*, 2025.

Best Paper & Oral, NeurIPS 2025 CCFM WS

- L. Le Boudec, E. de Bézenac, L. Serrano, R. D. Regueiro-Espino, **Y. Yin**, and P. Gallinari. Learning a neural solver for parametric PDE to enhance physics-informed methods. In *ICLR 2025*.
- A. Kassai Koupaï, J. Mifsut-Benet, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. GEPS: Boosting generalization in parametric PDE neural solvers through adaptive conditioning. In *NeurIPS 2024*.
- E. Le Naour, L. Serrano, L. Migus, **Y. Yin**, G. Agoua, N. Baskiotis, P. Gallinari, and V. Guigue. Time series continuous modeling for imputation and forecasting with implicit neural representations. *TMLR*, 2024.
- L. Serrano, L. Le Boudec, A. Kassai Koupaï, T. X. Wang, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. Operator learning with neural fields: Tackling PDEs on general geometries. In *NeurIPS 2023*.
- **Y. Yin**^{*}, M. Kirchmeyer^{*}, J.-Y. Franceschi^{*}, A. Rakotomamonjy, and P. Gallinari. Continuous PDE dynamics forecasting with implicit neural representations. In *ICLR 2023*. **Spotlight**
- M. Kirchmeyer^{*}, **Y. Yin**^{*}, J. Donà, N. Baskiotis, A. Rakotomamonjy, and P. Gallinari. Generalizing to new physical systems via context-informed dynamics model. In *ICML 2022*. **Spotlight**
- **Y. Yin**, I. Ayed, E. de Bézenac, N. Baskiotis, and P. Gallinari. LEADS: Learning dynamical systems that generalize across environments. In *NeurIPS 2021*.
- **Y. Yin**^{*}, V. Le Guen^{*}, J. Donà^{*}, E. de Bézenac^{*}, I. Ayed^{*}, N. Thome, and P. Gallinari. Augmenting physical models with deep networks for complex dynamics forecasting. In *ICLR 2021*.

Oral; also in J. Stat. Mech. Theory Exp.

- D. Huang, R.K. Zhang, **Y. Yin**, Y.D. Wang, and Y.H. Wang. Local feature approach to dorsal hand vein recognition by centroid-based circular key-point grid and fine-grained matching. *Image Vis. Comput.*, 2017.

Workshop papers

- **Y. Yin**, P. Khayatan, É. Zablocki, A. Boulch, and M. Cord. ReGentS: Real-world safety-critical driving scenario generation made stable.

In *ECCV 2024 Workshop on W-CODA*.

- L. Serrano, L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. INFINITY: Neural field modeling for reynolds-averaged navier-stokes equations. In *ICML 2023 Workshop on SynS & ML*.
- L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. Multi-scale physical representations for approximating PDE solutions with graph neural operators. In *ICLR 2022 Workshop on GTRL*.
- **Y. Yin**, A. Pajot, E. De Bézenac, and P. Gallinari. Unsupervised inpainting for occluded sea surface temperature sequences. In *CI 2019*.

Preprints

- **Y. Yin**, E. Ramzi, M. Lafon, V. Charraut, V. Bares, Y.-H. Xu, É. Zablocki, A. Boulch, T. Buhet, A. Bursuc, and M. Cord. Pictura: Perspective-view self-play at scale for driving. *arXiv* 2607.26005, 2026.
- Y.-H. Xu, É. Zablocki, **Y. Yin**, E. Ramzi, E. Kirby, A. Boulch, and M. Cord. TOAD: Test-time trajectory optimization for autonomous driving. *arXiv* 2606.07170, 2026.

Full list of publications on Google Scholar

Selected Talks

Oral, NeurIPS 2025 CCFM Workshop	Dec 2025
Poster, ECCV 2024	Sep 2024
Workshop on Mathematical Foundations of AI, DATAIA-SCAI	Jan 2024
Tutorial, ECML-PKDD 2023	Sep 2023
Seminar, Signal Processing Lab (LTS4), EPFL	May 2023
Spotlight, ICLR 2023	May 2023
AI4Science Talks, Univ. of Stuttgart & NEC Labs Europe	Apr 2023
Seminar, Criteo AI Lab	Nov 2022
Spotlight, ICML 2022	Jul 2022

Full list of talks on my website